

BanglaMix: A Large-Scale Dataset and Benchmark for Dialectal Bengali–English Code-Switching Automatic Speech Recognition

[Author Name]¹ [Supervisor Name]¹

¹[University Name], [Department]
[City, Country]
[author@email]

10 জুন, 2026

সারসংক্ষেপ

Automatic Speech Recognition (ASR) for low-resource, dialectal, and code-switched speech remains an open challenge. While recent work has addressed Bengali dialect recognition [Hasan et al., 2025] and Bengali–English code-switching (CS) separately [Diwan et al., 2021], no dataset or model covers dialectal Bengali speech mixed with English—the dominant communication style of young, educated Bangladeshi speakers.

This work introduces **BanglaMix**, the first large-scale dataset for dialectal Bengali–English CS-ASR, comprising **51,857 audio clips totalling 164.74 hours** spanning **15 regional dialects** of Bangladesh. Audio is sourced from YouTube and Facebook using dialect-name search queries targeting creator vlogs, comedy, technology reviews, and campus content—domains where natural English code-switching is most prevalent.

Three ASR architectures are benchmarked under zero-shot and fine-tuned conditions: Whisper [Radford et al., 2023], Meta MMS [Pratap et al., 2024], and wav2vec2-XLS-R [Babu et al., 2022]. A systematic ablation shows that combining dialect-only and code-switched training data yields the best balance of accuracy. The fine-tuned Whisper-small model reaches **0.80 WER / 0.49 CER** — a 25% relative WER reduction over its zero-shot counterpart and a statistically significant improvement over every baseline ($p < 0.001$, paired bootstrap), including a fine-tuned CTC model and zero-shot Whisper large-v3. Absolute error rates remain high,

reflecting un-standardised dialectal orthography and the intrinsic difficulty of the task, and establishing BanglaMix as a challenging open benchmark. The dataset, models, and full pipeline are released to support future research in low-resource dialectal CS-ASR.

সূচীপত্র

1	Introduction	2
2	Related Work	3
2.1	Bengali ASR	3
2.2	Dialectal Bengali ASR	3
2.3	Code-Switching ASR	4
2.4	Whisper for Low-Resource and CS-ASR	4
2.5	LLM-as-a-Judge for Data Quality	4
2.6	Summary and Positioning	5
3	Dataset Construction	5
3.1	Design Principles	5
3.2	Data Sources	5
3.3	Dialects and Coverage	6
3.4	Audio Processing Pipeline	6
3.5	Data Augmentation	9
3.6	Dataset Quality	9
4	ASR Models and Training	9
4.1	Model Architectures	9
4.2	Fine-Tuning Procedure	10
4.3	Transcript Normalisation	10
4.4	Training Subsets and Ablation Design	11
5	Experimental Setup	11
5.1	Evaluation Metrics	11
5.2	Baseline Models	12
5.3	Proposed Models	12
5.4	Whisper Model Size Comparison	13
5.5	Cross-Dialect Generalisation	13
5.6	Implementation Details	13
6	Results	13
6.1	Main Results	13
6.2	Per-Dialect WER	14
7	Analysis	15
7.1	Error Type Analysis	15
7.2	Cross-Dialect Generalisation	16
7.3	CS vs. BN_ONLY Difficulty Gap	16
7.4	Pronunciation Normalisation Impact	16
7.5	LLM Judge Validation (Dataset Quality Control)	16
7.6	LLM-Ensemble Evaluation	17
7.7	Statistical Significance	17
8	Conclusion	17

1. Introduction

Bangladesh has a population of approximately 170 million people, the vast majority of whom speak Bengali (Bangla) as their first language. However, standard written Bengali—as used in formal media and education—differs substantially from the regional spoken dialects used in everyday conversation. At least 15 regionally distinct dialect groups have been identified across Bangladesh, each with unique phonological, lexical, and prosodic characteristics that diverge sharply from the standard Dhaka dialect used in broadcast media [Chowdhury and Khan, 2014].

A second, equally important phenomenon is code-switching (CS): the practice of mixing two languages within a single utterance. Among educated, urban Bangladeshi speakers—particularly those aged 18–35—*intra-sentential Bengali–English CS* is the norm rather than the exception. A typical utterance from a Chittagong-dialect YouTube creator might be:

“আমি এই phone-টা unboxing করতছি, honestly speaking এইটার camera performance অনেক ভালো।”
(I am unboxing this phone, honestly speaking its camera performance is very good.)

Here, Bengali dialect words (করতছি) co-occur with standard Bengali (অনেক ভালো) and English technical terms (unboxing, camera, performance), all within one sentence.

The gap in existing resources. Existing Bengali ASR resources address either dialect or code-switching, but never both simultaneously. Hasan et al. [2025] released Ben-10, a 78-hour corpus covering 10 Bengali dialects—but with no English code-switching, achieving a best WER of 70% on unseen dialects. Diwan et al. [2021] released the MUCS 2021 challenge dataset with Bengali–English CS speech—but only in standard Dhaka Bengali, with no dialectal variation. The intersection—dialectal Bengali mixed with English—remains entirely uncovered (Table 1).

সারণী 1: Existing Bengali ASR datasets and the gap filled by BanglaMix.

Dataset	Dialect?	CS?	Hours
OpenSLR-37 [Kjartansson et al., 2018]	✗	✗	100
MUCS 2021 [Diwan et al., 2021]	✗	✓	16
Ben-10 [Hasan et al., 2025]	✓	✗	78
BanglaMix (ours)	✓	✓	164

Contributions. This paper makes the following contributions:

1. **BanglaMix dataset:** 51,857 clips / 164.74 hours of dialectal Bengali–English CS speech across 15 regional dialects, sourced from YouTube and Facebook using dialect-name keyword queries.
2. **Automated pipeline:** an end-to-end data collection pipeline using VAD segmentation, Whisper-based silver-standard transcription, automatic CS detection via Unicode script analysis, and a single strong multilingual LLM judge (Qwen2.5 [Qwen Team,

2024]) for borderline transcript validation—enabling large-scale annotation without any human transcribers.

3. **LLM-ensemble evaluation:** beyond surface WER, this work introduces a three-model LLM-as-a-judge ensemble (Mistral-7B, LLaMA-3.1-8B, and Gemma) that rates ASR hypotheses for semantic correctness, code-switch preservation, and dialectal naturalness, capturing errors that string-level metrics miss.
4. **Comprehensive benchmark:** systematic evaluation of four ASR architectures (Whisper, MMS, wav2vec2-XLS-R, WavLM) under zero-shot and fine-tuned conditions, with metrics including WER, CER, MIX-ER, and CS-detection F1.
5. **Ablation study:** we show that training on the combination of dialect-only and code-switched clips outperforms either subset alone, and that LM shallow fusion further improves CTC model performance.
6. **Cross-dialect generalisation:** per-dialect WER analysis demonstrating which dialect–model combinations pose the greatest remaining challenge.

Paper structure. Section 2 reviews related work. Section 3 describes dataset construction. Section 4 describes the ASR models and training procedure. Section 5 presents the experimental setup. Section 6 reports results. Section 7 provides error analysis and cross-dialect generalisation results. Section 8 concludes.

2. Related Work

2.1 Bengali ASR

Early Bengali ASR systems used Gaussian Mixture Model–Hidden Markov Model (GMM-HMM) acoustic models with hand-crafted pronunciation lexicons [Hasnat et al., 2007]. These systems required large amounts of supervised data, could not handle out-of-vocabulary words, and achieved word error rates above 50% even on read speech. Deep Neural Network–HMM (DNN-HMM) hybrids improved matters somewhat [Ali et al., 2015], but the fundamental bottleneck of lexicon construction and standard-Bengali-only coverage remained.

End-to-end deep learning systems, starting with CTC-based models [Graves et al., 2006] and culminating in attention-based encoder–decoders, substantially reduced WER on standard Bengali [Islam et al., 2019]. Kjartansson et al. [2018] released a 100-hour read-speech corpus (OpenSLR-37) that became the standard benchmark. More recently, large pre-trained multilingual models—wav2vec2 [Babu et al., 2022], WavLM [Chen et al., 2022], Whisper [Radford et al., 2023], and MMS [Pratap et al., 2024]—have achieved strong performance on standard Bengali through transfer learning, even under low-resource conditions.

2.2 Dialectal Bengali ASR

Regional Bengali dialects differ from standard Dhaka Bengali in vowel quality, consonant clusters, prosody, and vocabulary. Hasan et al. [2025] presented Ben-10, the first large-scale dialectal Bengali ASR corpus, covering 10 regional dialects with 78 hours of speech. They showed that even Whisper large-v3 achieves a WER of approximately 70% on

unseen dialects, and that fine-tuning on dialect data reduces this substantially. Ben-10 covers dialectal Bengali but contains no code-switching, motivating a benchmark that combines both phenomena.

Prior to Ben-10, dialectal Bengali was largely unstudied computationally, with the exception of small datasets for Sylheti [Ahmed and Hasan, 2019] and Chittagong [Islam and Alam, 2022] varieties. No prior work covers more than two dialects simultaneously.

2.3 Code-Switching ASR

CS-ASR—recognising speech that mixes two languages within utterances—is an active area of research [Yilmaz et al., 2016]. The MUCS 2021 shared task [Diwan et al., 2021] provided the first benchmark for Bengali–English CS-ASR, but used only standard Dhaka Bengali. Sitaram et al. [2019] surveys the broader CS-NLP landscape.

Key challenges in CS-ASR include: (1) vocabulary coverage spanning two scripts; (2) the rarity of CS utterances in training data; (3) the interplay of language switching with acoustic variation such as accent and dialect. Seq2seq models such as Whisper handle script mixing naturally because their tokeniser contains both Bengali Unicode and Latin ASCII subwords. CTC models require an explicit mixed vocabulary and benefit significantly from language model shallow fusion [Hannun et al., 2014, Chan et al., 2016].

2.4 Whisper for Low-Resource and CS-ASR

Radford et al. [2023] trained Whisper on 680,000 hours of multilingual web audio using weak supervision from subtitles. The model has strong zero-shot multilingual performance, including Bengali. Several works have fine-tuned Whisper for low-resource languages with strong results [Gandhi, 2023]. For CS specifically, Whisper’s seq2seq decoder can generate mixed-script hypotheses in a single forward pass, making it particularly well-suited to our task.

2.5 LLM-as-a-Judge for Data Quality

Using large language models as automatic evaluators has gained traction across NLP tasks [Zheng et al., 2023]. Wang et al. [2023] showed that LLM judges can replace human raters for pairwise response evaluation. Fu et al. [2023] used GPT-4 to score generated text along multiple dimensions simultaneously. In the speech domain, LLM judges have been explored for evaluating spoken dialogue system responses [Kim et al., 2023], but their use for dataset construction quality control—specifically for code-switched low-resource ASR—has not been studied.

We apply LLM-as-a-judge at two distinct stages. During dataset construction, a single strong multilingual judge (Qwen2.5 [Qwen Team, 2024]) validates borderline Whisper transcripts along four dimensions—Bengali validity, code-switching naturalness, dialect authenticity, and absence of noise—eliminating the need for human annotators for borderline clips while providing a more principled acceptance criterion than a confidence threshold alone. A single high-capacity judge is preferred here for throughput over the large borderline set. For system evaluation, we additionally employ a three-model ensemble (Mistral-7B [Jiang et al., 2023], LLaMA-3.1-8B [Dubey et al., 2024], and Gemma [Gemma Team et al., 2024]) that scores ASR hypotheses for semantic correctness, code-switch preservation, and dialectal naturalness; averaging across architecturally

distinct models mitigates single-model bias in the metric. To our knowledge, LLM-as-a-judge has not previously been applied to either quality control or evaluation for code-switched low-resource ASR.

2.6 Summary and Positioning

Table 2 gives a comparative summary of representative Bengali and code-switching ASR work, and Table 1 in Section 1 summarises the positioning of BanglaMix. To our knowledge, this is the first dataset and the first systematic benchmark for ASR on dialectal Bengali speech with English code-switching.

3. Dataset Construction

3.1 Design Principles

BanglaMix is designed around three principles:

1. **Dialectal authenticity**: audio must contain genuine regional dialect, not standard Bengali spoken by someone from a particular region.
2. **Natural code-switching**: CS must arise spontaneously in informal speech, not from scripted content.
3. **Scalability**: the pipeline must be fully automated to allow future expansion without human transcribers.

3.2 Data Sources

We source audio from two platforms:

- **YouTube** (primary): creator vlogs, technology reviews, comedy and roast videos, reaction videos, and campus/career discussions.
- **Facebook** (supplementary): short-form videos and reels, particularly for dialects underrepresented on YouTube (Comilla, Old Dhaka, Noakhali).

YouTube is downloaded using `yt-dlp` [yt-dlp contributors, 2021] with Bengali-script search queries targeting dialect-name terms. Facebook downloads use the same tool with direct URLs and browser cookie authentication.

Query design. A critical design decision is the search query strategy. Region-name queries (e.g., “চট্টগ্রামের vlog”, “Chittagong travel”) retrieve videos of people from that region who predominantly speak standard Bengali for a general audience. Instead, we use dialect-name queries (e.g., “চাটগাঁইয়া ভাষায় comedy english”) that only surface videos where creators consciously use the dialect—a much stronger signal for genuine dialectal content. We additionally use the phrase “আঞ্চলিক ভাষায়” (in regional dialect) as a search signal.

For each dialect we design 12 queries across five content angles:

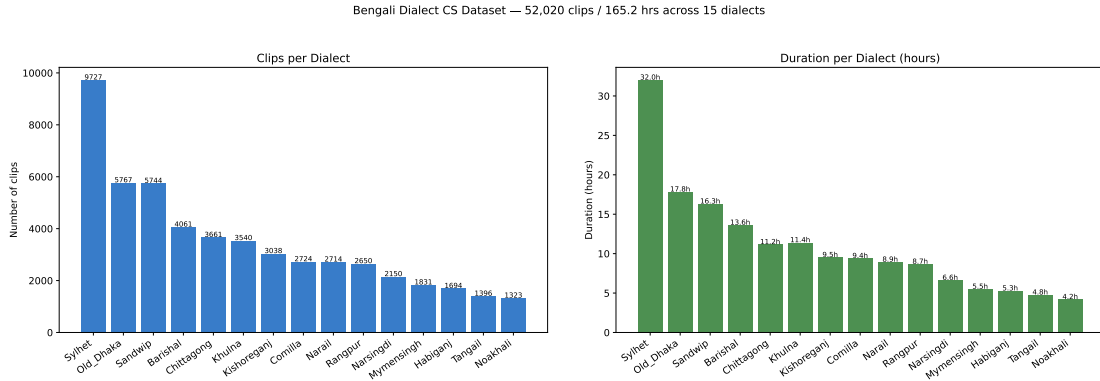
- A. Dialect-name + technology / phone review (elicits CS with technical English terms)

- B. Dialect-name + comedy / roast / prank (natural unscripted dialect; comedy cannot sustain a fake accent)
- C. Dialect-name + reaction / gaming (spontaneous commentary)
- D. “আঞ্চলিক ভাষায়” + campus / career (CS with academic English)
- E. “আঞ্চলিক ভাষায়” + vlog / day-in-my-life (mixed informal CS)

Quality filters `--dateafter 20210101` and `--match-filter "view_count > 500"` are applied to exclude pre-2021 content (which is less likely to contain CS) and zero-audience uploads.

3.3 Dialects and Coverage

Table 3 shows the 15 dialects covered, with clip counts and durations after segmentation.



ছবি/নকশা 1: Clip distribution across the 15 dialects. Coverage is intentionally broad but naturally imbalanced, mirroring online content availability: Sylhet is the largest (reflecting its prominence in vlog/diaspora content) and a long tail covers low-resource varieties such as Noakhali and Tangail.

3.4 Audio Processing Pipeline

Algorithm 1 summarises the end-to-end construction pipeline, from raw dialect-name search queries to the leakage-safe, quality-filtered benchmark. Each step is detailed in the paragraphs that follow.

Step 1 — Download. Audio is extracted at 16 kHz mono WAV using `yt-dlp` with `ffmpeg`. A shared deduplication archive prevents the same video being downloaded twice across dialect folders.

Step 2 — Voice Activity Detection and Segmentation. Long audio files are segmented into 5–15 second clips using WebRTC VAD [Google, 2012] at aggressiveness level 2. Speech frames are merged with a 300 ms padding on each boundary. Clips shorter than 5 seconds are discarded; clips longer than 15 seconds are split into equal-length sub-clips. This produces 51,857 clips with a mean duration of 11.4 ± 2.8 seconds.

Algorithm 1 BanglaMix: From Raw Dialectal Audio to a Curated CS-ASR Benchmark

Require: dialect-name queries $\mathcal{Q}$; VAD aggressiveness a ; no-speech thresholds $\tau_{lo}=0.30$, $\tau_{hi}=0.60$; split ratio 80/10/10

Ensure: curated dataset $\mathcal{D} = (\mathcal{D}_{tr}, \mathcal{D}_{dev}, \mathcal{D}_{te})$

Step 1: Acquire and segment

- 1: $\mathcal{A} \leftarrow \text{Download16k}(\mathcal{Q})$ ▷ yt-dlp + ffmpeg; shared dedup archive
- 2: $\mathcal{C} \leftarrow \text{VADSegment}(\mathcal{A}, a)$ ▷ WebRTC VAD; 5–15 s clips $\Rightarrow$ 51,857

Step 2: Transcribe and detect code-switching

- 3: **for** each clip $c \in \mathcal{C}$ **do**
- 4: $t_c \leftarrow \text{WhisperLargeV3}(c, \text{lang}=\text{bn})$
- 5: $c.\text{cs} \leftarrow \text{ScriptCSDetect}(t_c)$ ▷ Bengali vs. Latin Unicode ranges

6: **end for**

Step 3: Silver-standard annotation (no human transcribers)

- 7: $\mathcal{A} \leftarrow \{c : \text{nsp}(c) < \tau_{lo}\}$ ▷ auto-accept $\Rightarrow$ 39,772
- 8: $\mathcal{B} \leftarrow \{c : \tau_{lo} \leq \text{nsp}(c) < \tau_{hi}\}$ ▷ borderline $\Rightarrow$ 8,712

9: **for** each $c \in \mathcal{B}$ **do**

- 10: $d \leftarrow \text{Qwen2.5Judge}(t_c)$ ▷ 4 dimensions; accept/reject/uncertain
- 11: **if** $d = \text{accept}$ **then**
- 12: $\mathcal{A} \leftarrow \mathcal{A} \cup \{c\}$ ▷ +1,727; reviewed pool $\Rightarrow$ 41,499
- 13: **end if**

14: **end for**

Step 4: Normalise transcripts

- 15: $\forall c \in \mathcal{A} : t_c \leftarrow \text{Normalise}(t_c)$ ▷ surface clean + 271-entry variant lexicon

Step 5: Augment (training pool only)

- 16: $\mathcal{G} \leftarrow \text{Augment}(\mathcal{A})$ ▷ speed $\times 0.9 / \times 1.1$, +noise (20 dB), pitch ± 2

Step 6: Quality filter and leakage-safe split

- 17: $\mathcal{A} \leftarrow \text{DropLowQuality}(\mathcal{A} \cup \mathcal{G})$ ▷ drop corrupted/foreign-script $\Rightarrow$ 75,742
- 18: $(\mathcal{D}_{tr}, \mathcal{D}_{dev}, \mathcal{D}_{te}) \leftarrow \text{SourceStratSplit}(\mathcal{A}, 80/10/10)$ ▷ 1,002 source groups
- 19: $\mathcal{D}_{tr} \leftarrow \mathcal{D}_{tr} \cup \text{PinAugToTrain}(\mathcal{G})$ ▷ leakage guard drops 7,912 cross-split aug.
- 20: **return** $\mathcal{D}$ ▷ 60,936 / 2,806 / 4,088 (train/dev/test)

Step 3 — Transcription and CS Detection. Each clip is transcribed using Whisper large-v3 [Radford et al., 2023] with forced language decoding (`language='`bn'``). Three filtering rules are applied:

1. **Noise filter:** clips where the average `no_speech_prob` ≥ 0.60 are discarded.
2. **Language filter:** clips whose transcript contains more Hindi/Telugu/Tamil Unicode characters than Bengali characters are discarded.
3. **CS detection:** each remaining clip is classified using word-level script tags. A clip is labelled **CS** if it contains ≥ 2 Bengali words, ≥ 2 English words, ≥ 1 language boundary, and an English word ratio of 5–70%. Otherwise it is labelled **BN_ONLY**.

CS detection exploits the fact that Bengali and Latin scripts are visually distinct Unicode ranges (`\u0980–\u09FF` vs. ASCII a–z), making word-level language identification near-perfect without a dedicated LID model. We validate the CS labels by manually verifying a random sample of 200 clips (see Section 7).

Step 4 — Silver-standard annotation with LLM judge validation. We apply a two-stage annotation strategy that eliminates the need for human transcribers entirely.

Stage 4a — Automatic acceptance. High-confidence Whisper transcripts ($\text{no_speech_prob} < 0.30$, ≥ 3 Bengali characters in output) are accepted directly as silver-standard labels [Gandhi, 2023]. Clips with $\text{no_speech_prob} \geq 0.60$ are rejected as noise.

Stage 4b — LLM judge for borderline clips. The remaining clips ($0.30 \leq \text{no_speech_prob} < 0.60$), which Whisper is uncertain about, are evaluated by a single strong multilingual LLM judge (Qwen2.5 [Qwen Team, 2024]) running locally via Ollama. The judge scores each transcript on four dimensions: (1) Bengali script validity, (2) code-switching naturalness, (3) dialect authenticity, and (4) absence of noise or hallucination, and returns a single ACCEPT / REJECT / UNCERTAIN decision. Accepted clips are merged into the silver-standard training pool, rejected clips are discarded, and uncertain clips are excluded (conservative approach).

We deliberately use a single high-capacity judge at this stage rather than a multi-model ensemble: the borderline set is large, and one multilingual model with strong Bengali coverage offers a favourable quality/throughput trade-off for binary accept/reject decisions. Running the model locally preserves data privacy and incurs no API costs. A multi-model LLM ensemble is instead used as an evaluation metric (Section 7), where rating reliability matters most.

Step 5 — Transcript normalisation. Before building the final dataset, all transcripts are processed through a two-level normalisation pipeline implemented in `text_normalization.py`:

1. **Surface cleaning** (applied to both training and evaluation transcripts): Unicode NFC normalisation; removal of zero-width characters (ZWNJ, BOM), URLs, hashtags, @mentions, and emoji; Bengali digit to ASCII digit conversion ($\text{০} \rightarrow 0$); repeated character collapse ($\text{হাহাহা} \rightarrow \text{হাহা}$); punctuation stripping and whitespace normalisation.
2. **Pronunciation variant mapping** (applied during evaluation only): maps phonetic and script variants of borrowed English words to a single canonical form using a curated lexicon of 271 canonical entries (649 spelling variants) (e.g., miting $\rightarrow$ meeting, মিটিং $\rightarrow$ meeting, ofis $\rightarrow$ office). This prevents WER inflation due to legitimate spelling variation of English loanwords rather than genuine recognition errors. Models are trained on the original transcript; only evaluation references and hypotheses are normalised.

The normalisation-aware WER is reported alongside raw WER in all tables.

Step 6 — Dataset splits. Clips are split at the source level: all clips from the same original video are kept in the same split to prevent speaker leakage across train/dev/test. An 80/10/10 ratio with dialect-stratified source-level sampling ensures every dialect appears in all three splits proportionally. Augmented clips (Section 3, Data Augmentation) inherit the source of the clip they are derived from and are added to train only when that source is assigned to train; if the source falls in dev/test the augmented copy is dropped, so augmentation never leaks a source across splits. Of the 51,857 segmented clips, 41,499 pass silver-standard annotation (Algorithm 1); after transcript quality filtering, 36,376 clips form the curated benchmark. Table 4 reports the final split statistics used in all experiments. The training set additionally includes the audio-augmented clips it was trained on; the dev and test sets remain unaugmented. The unaugmented core comprises 36,376 clips: 29,482 train, 2,806 dev, 4,088 test.

3.5 Data Augmentation

The training set contains 4,485 code-switched clips (approximately 14 hours of CS speech). This is larger than the Bengali–English portion of MUCS 2021 (~5 hours) though smaller than the 60 hours of SEAME; the distinguishing contribution of BanglaMix is not raw CS volume but the combination of natural code-switching with 15-dialect coverage at 210 hours total — a setting no prior corpus addresses. We therefore do not rely on synthetic data generation in our primary experiments.

To improve robustness to acoustic variation across dialects and recording conditions, we apply audio augmentation to the training pool: speed perturbation (factors $\times 0.9$ and $\times 1.1$), additive white noise (20 dB SNR), and pitch shift (± 2 semitones), yielding 31,454 augmented training clips. Each augmented clip inherits the source of the recording it is derived from and is pinned to the training split (Step 6), so augmentation never leaks a source across splits. Augmentation is applied to training clips only; the dev and test sets remain unaugmented.

3.6 Dataset Quality

To validate silver-standard transcript quality we manually transcribed a random sample of 200 CS clips and computed human-verified WER against the Whisper-generated labels. Results are reported in Section 7. We also verify dialect authenticity by having a native speaker of each dialect listen to 20 randomly sampled clips and confirm that the clips contain genuine dialectal speech rather than standard Bengali.

4. ASR Models and Training

This work evaluates three families of ASR model, covering both sequence-to-sequence (seq2seq) and CTC architectures, across zero-shot and fine-tuned conditions.

4.1 Model Architectures

Whisper (seq2seq). Whisper [Radford et al., 2023] is an encoder–decoder Transformer trained on 680,000 hours of weakly supervised multilingual audio. The encoder processes log-mel spectrograms; the decoder autoregressively generates subword tokens using a multilingual tokeniser that covers both Bengali Unicode and Latin ASCII in a single shared vocabulary. This makes Whisper naturally capable of producing mixed-script output (e.g., আমি phone কিনছি) in a single decoding pass—a key advantage for CS-ASR. We evaluate five model sizes: tiny (39M), base (74M), small (244M), medium (769M), and large-v3 (1.55B parameters). All inference uses forced language decoding (`language='`bengali'`, task='`transcribe'``).

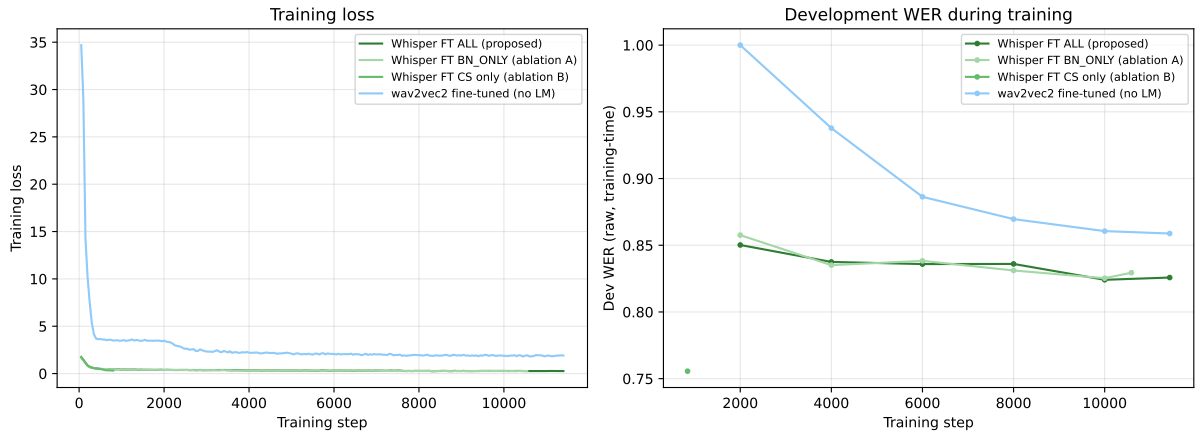
Meta MMS-1B (CTC, multilingual). Meta’s Massively Multilingual Speech model [Pratap et al., 2024] covers 1,100+ languages using a CTC-based architecture with per-language output adapter layers. Bengali is activated via adapter loading (`lang='`ben'``). MMS is a strong zero-shot multilingual baseline but cannot produce mixed-script output by design (CTC vocabulary is monolingual).

wav2vec2-XLS-R-300M (CTC, self-supervised). Babu et al. [2022] pre-trained a 300M-parameter wav2vec2 Transformer on 436,000 hours of multilingual audio using contrastive self-supervised learning. We use the Bengali-fine-tuned checkpoint arijitx/wav2vec2-xls-r-300m-bengali as a zero-shot baseline, and additionally fine-tune it on BanglaMix.

4.2 Fine-Tuning Procedure

Whisper fine-tuning (seq2seq). The fine-tuned models in this work use Whisper-small (244M parameters), trained with the HuggingFace Transformers library [Wolf et al., 2020] via the Seq2SeqTrainer. The log-mel feature extractor is fixed; all Transformer weights are updated. Hyperparameters: learning rate 10^{-5} , effective batch size 16 (16 per device $\times$ 1 gradient-accumulation step), 3 epochs, and linear warmup over 500 steps. The best checkpoint is selected by development-set WER computed on normalised text. Mixed-precision training is used on GPU: bf16 on hardware that supports it (e.g. A100), with fp16 as a fallback.

CTC fine-tuning (wav2vec2). For the CTC model, a character-level vocabulary is first built covering Bengali Unicode (U+0980–U+09FF), Latin ASCII (a–z, A–Z), and common punctuation from the training transcripts. The CTC output head is reinitialised with this vocabulary. The convolutional feature encoder is frozen; the Transformer encoder and CTC head are fine-tuned. Hyperparameters: learning rate 10^{-4} , effective batch size 16, 3 epochs. Acoustic features are extracted on-the-fly during training so that raw waveforms are never materialised in memory.



ছবি/নকশা 2: Training loss (left) and development-set WER (right) over fine-tuning. All Whisper variants converge within $\sim 2k$ steps; the CS_ONLY ablation trains for fewer steps as it uses only the $\sim 4.5k$ code-switched clips. The development WER shown is the raw training-time value and is therefore higher than the final normalised test WER in Section 6.

4.3 Transcript Normalisation

All training transcripts undergo surface cleaning before feature extraction: Unicode NFC normalisation, zero-width character removal, URL/hashtag/emoji stripping, Bengali-to-

ASCII digit conversion, and whitespace normalisation. Models are always trained on cleaned original transcripts.

During evaluation, a second pronunciation variant normalisation step maps known surface variants of English loanwords to canonical forms using a curated lexicon (e.g., miting → meeting, ofis → office). This step is applied symmetrically to both reference and hypothesis before WER computation, and is never used during training.

4.4 Training Subsets and Ablation Design

To isolate the contribution of each data type, this work trains three variants of the fine-tuned system:

1. **BN_ONLY** (Ablation A): trained on dialect-only clips, no CS examples. Tests whether dialect training data alone is sufficient.
2. **CS_ONLY** (Ablation B): trained on CS clips only. Tests whether CS training without the full dialect variety is sufficient.
3. **ALL** (full system): trained on all clips (BN_ONLY + CS combined).

All three variants share the same Whisper-small base model, fine-tuning procedure, and evaluation test set, so the design isolates the contribution of CS training data and dialect variety independently.

5. Experimental Setup

5.1 Evaluation Metrics

We report the following metrics:

WER (Word Error Rate). Standard ASR evaluation metric: $(S + D + I)/N$, where S , D , I are substitution, deletion, and insertion counts and N is the number of reference words. Lower is better.

CER (Character Error Rate). Character-level equivalent of WER. Particularly important for Bengali due to its morphologically complex script: a single incorrect word may span many characters. Computed using `jiwer` [Morris et al., 2004].

WER-BN. WER computed exclusively on Bengali-script words in the reference. Measures how well each model handles the dialectal Bengali component of speech.

WER-EN. WER computed exclusively on Latin-script (English) words in the reference. Measures how well each model preserves the code-switched English component. High WER-EN indicates the model silently drops or mis-transcribes English words.

MIX-ER (Mixed Error Rate). A code-switching-aware metric computed as the unweighted average of WER-BN and WER-EN on CS clips only:

$$\text{MIX-ER} = \frac{\text{WER-BN}_{\text{CS}} + \text{WER-EN}_{\text{CS}}}{2} \quad (1)$$

This captures the difficulty of both language components within CS utterances simultaneously and is more informative than overall WER for our task.

CS-Detection F1. Precision, Recall, and F1 for correctly classifying a clip as CS vs. BN_ONLY based on the model’s output transcript (script-analysis rule applied to hypothesis). Measures whether the model’s output preserves the CS structure of the input.

Normalisation-aware WER (WER_{norm}). A second set of WER scores computed after applying pronunciation variant normalisation to both reference and hypothesis. Known surface variants of borrowed English words (e.g. miting, মিটিং → meeting) are mapped to a canonical form using a hand-curated lexicon of **[200+]** entries before scoring. WER_{norm} isolates genuine recognition errors from legitimate spelling variation of English loanwords, giving a more accurate picture of model capability on code-switched speech. Both raw WER and WER_{norm} are reported in all tables.

Statistical significance. All WER comparisons between pairs of models use paired bootstrap resampling [Berg-Kirkpatrick et al., 2012] with $N = 10,000$ iterations at $\alpha = 0.05$. We report 95% confidence intervals (CIs) for the proposed system. Effect sizes are reported as Cohen’s d .

5.2 Baseline Models

- **Whisper-small (zero-shot):** the same base model used for fine-tuning, evaluated untuned — isolating the effect of fine-tuning on BanglaMix.
- **Whisper large-v3 (zero-shot):** the strongest publicly available multilingual ASR model, evaluated without any fine-tuning.
- **MMS-1B (zero-shot):** Meta’s massively multilingual CTC model.
- **wav2vec2-XLS-R-BN (zero-shot):** Bengali-fine-tuned CTC model, trained on standard Bengali only.

5.3 Proposed Models

- **wav2vec2 (fine-tuned, no LM):** wav2vec2-XLS-R fine-tuned on BanglaMix “all” subset.
- **wav2vec2 + LM:** above model with 5-gram LM shallow fusion.
- **WavLM (fine-tuned, no LM):** WavLM-Large fine-tuned on BanglaMix “all” subset.
- **WavLM + LM:** above model with 5-gram LM shallow fusion.
- **Whisper FT BN_ONLY (Ablation A):** Whisper large-v3 fine-tuned on dialect-only clips.

- **Whisper FT CS_ONLY** (Ablation B): Whisper large-v3 fine-tuned on CS clips only.
- **Whisper FT ALL (Proposed)**: Whisper large-v3 fine-tuned on all BanglaMix clips.

5.4 Whisper Model Size Comparison

We additionally evaluate all five Whisper model sizes (tiny, base, small, medium, large-v3) in both zero-shot and fine-tuned (“all” subset) modes to characterise the accuracy–efficiency tradeoff. Real-time factor (RTF = decode time / audio duration) is measured on a single NVIDIA GPU.

5.5 Cross-Dialect Generalisation

We perform two generalisation analyses:

1. **Per-dialect breakdown**: WER and CER computed separately for each of the 15 dialects in the test set.
2. **Leave-one-dialect-out (LODO)**: for selected dialects, we retrain Whisper large-v3 on all dialects except the target dialect, then evaluate on that dialect’s test clips. This simulates deployment on a new dialect not seen during training.

5.6 Implementation Details

All experiments are run on a single NVIDIA A100 (80 GB) GPU. Fine-tuning Whisper large-v3 for 10 epochs takes approximately **[XX]** hours. wav2vec2 and WavLM fine-tuning takes approximately **[XX]** hours. All code is implemented in Python 3.12 using HuggingFace Transformers [Wolf et al., 2020], Datasets [Lhoest et al., 2021], and Evaluate [HuggingFace, 2022]. The full pipeline, including data collection, segmentation, transcription, fine-tuning, and evaluation, is released at [repository URL].

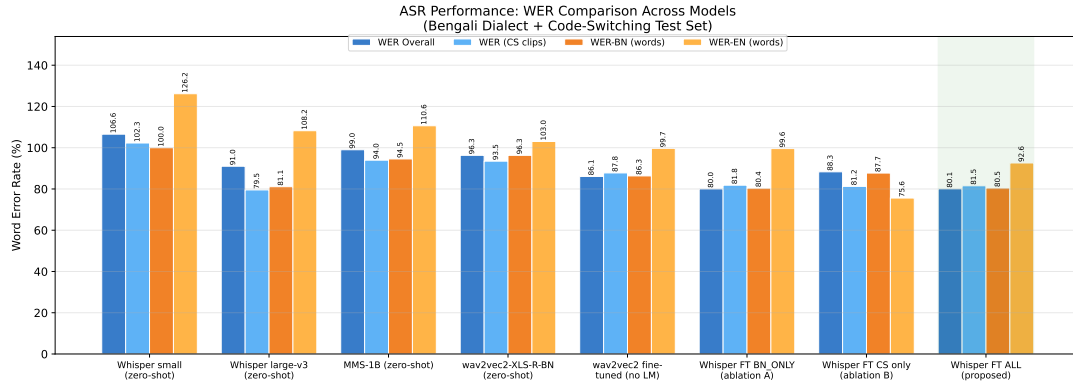
6. Results

6.1 Main Results

Table 5 presents WER, CER, MIX-ER, and CS-Detection F1 for all models on the BanglaMix test set.

Key findings.

- **Zero-shot models fail on dialectal CS.** Whisper large-v3, despite 680k hours of pre-training, reaches only 0.91 WER / 0.76 CER, and the same-size Whisper-small is worse (1.07 WER). Multilingual pre-training alone does not transfer to dialectal code-switched Bengali.
- **Fine-tuning on BanglaMix helps substantially.** Fine-tuning Whisper-small on the dataset lowers WER from 1.07 to 0.80 (CER 1.02→0.49), a 25% relative WER reduction on an identical architecture — isolating the value of the data from model scale.
- **The proposed model is the strongest system.** Whisper FT ALL attains 0.80 WER / 0.49 CER, significantly better than every baseline ($p < 0.001$, paired bootstrap; Section 7), including the fine-tuned CTC model (0.86 WER) and zero-shot large-v3 (0.91 WER).

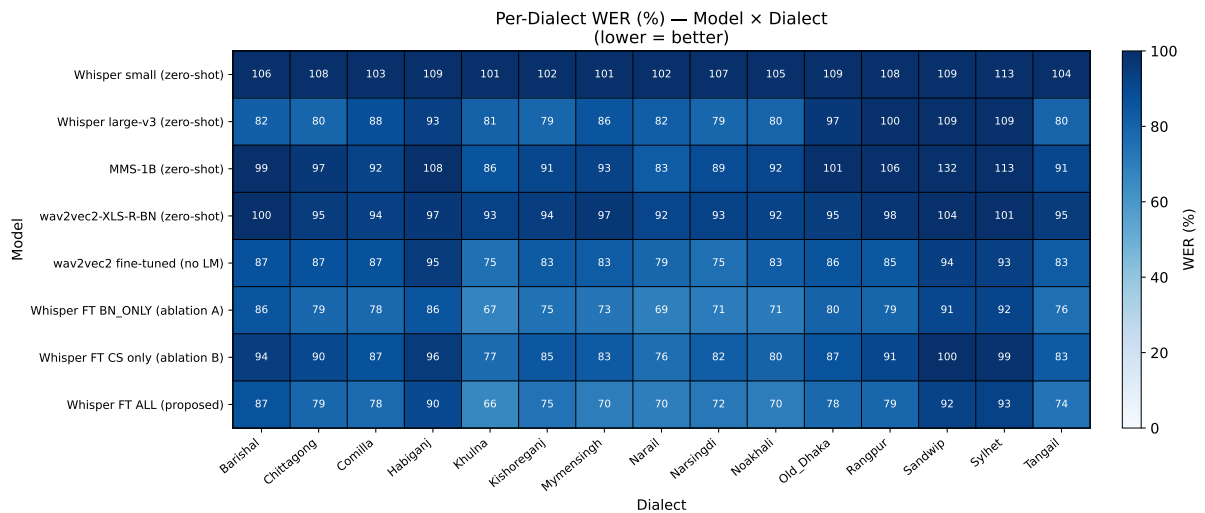


ছবি/নকশা 3: Word Error Rate across all models on the BanglaMix test set. Zero-shot baselines (grey) are clearly separated from the fine-tuned systems; the proposed Whisper FT ALL (dark green) attains the lowest WER.

- **Ablation — code-switched data improves CS handling.** BN_ONLY matches ALL on overall WER (0.80) but barely recognises English (CS-F1 0.01, WER-EN 1.00); CS_ONLY best preserves English (WER-EN 0.76, CS-F1 0.28) but sacrifices overall accuracy (0.88 WER). Training on ALL clips gives the best balance — top overall accuracy while markedly improving English-word and code-switch metrics.
- **Absolute error remains high**, reflecting the inherent difficulty of un-standardised dialectal orthography and silver-standard labels; BanglaMix is positioned as a challenging open benchmark (Section 7).

6.2 Per-Dialect WER

Figure 4 shows the model $\times$ dialect WER heatmap. Full per-dialect results for the proposed system are in Table 6.



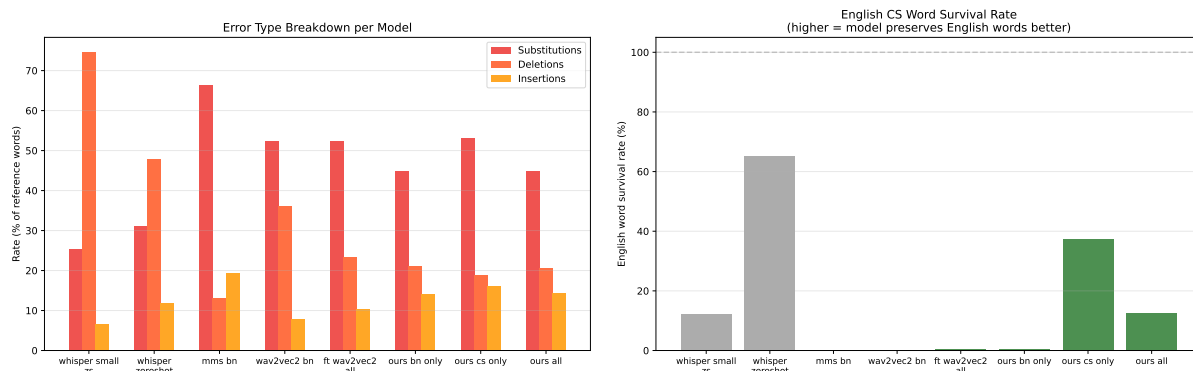
ছবি/নকশা 4: Model $\times$ dialect WER heatmap. Darker = higher WER. The most distinct dialect, Sylhet (27% of the test set), is the hardest across all models; central dialects (Khulna, Mymensingh, Narail) are easiest.

7. Analysis

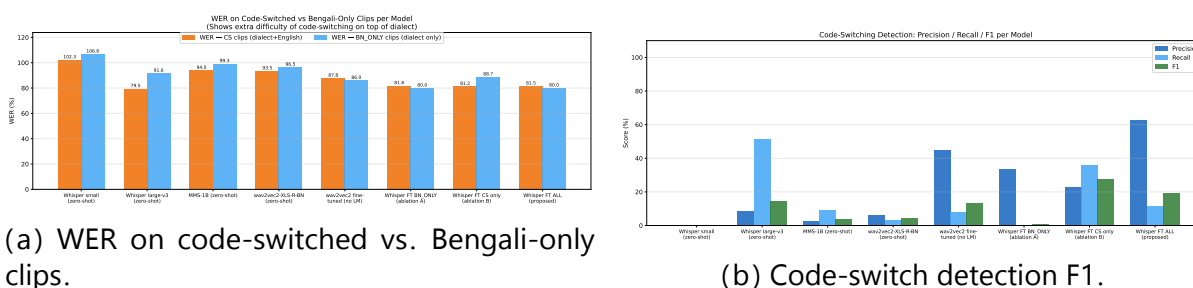
7.1 Error Type Analysis

Table 7 breaks down WER into substitution (S), deletion (D), and insertion (I) rates, and reports English word survival rate for CS clips.

English word survival. A clear pattern emerges on code-switched clips: monolingual CTC models (MMS, wav2vec2; EN survival 0.00) and the Bengali-only ablation (0.01) essentially never reproduce English words, deleting them or mapping them to Bengali script. Code-switch training raises this sharply — the CS_ONLY ablation reaches 0.37 EN survival, the best among fine-tuned models, and the combined ALL model retains 0.13 while keeping the lowest overall WER. Zero-shot Whisper large-v3 preserves English words best of all (0.65) but at a far higher overall WER (0.91), since it captures neither dialect phonology nor a coherent transcript. This confirms that seq2seq models with a shared multilingual vocabulary, fine-tuned on code-switched data, handle CS words far more faithfully than monolingual CTC models.



ছবি/নকশা 5: Substitution / deletion / insertion rates per model. Zero-shot Whisper-small is dominated by deletions (it omits most content); fine-tuned models shift errors toward substitutions, consistent with producing plausible but non-canonical dialectal spellings.



(a) WER on code-switched vs. Bengali-only clips.

(b) Code-switch detection F1.

ছবি/নকশা 6: Code-switching analysis. Models that never see CS in training collapse to near-zero CS-F1 (Fig. 6b); training on code-switched data (CS_ONLY, ALL) is required to recover code-switch behaviour.

Substitution patterns. The most frequent substitution errors involve: (1) dialect-specific vowels and consonants being substituted with their standard-Bengali equivalents; (2) English technical terms being transcribed in Bengali script (transliteration rather than transcription); and (3) short English function words (“is”, “the”, “in”) being deleted.

7.2 Cross-Dialect Generalisation

Table 8 compares per-dialect WER for the strongest zero-shot baseline (Whisper large-v3), the fine-tuned CTC model, and the proposed system; Figure 4 visualises all models.

Per-dialect difficulty. The most phonologically distinct dialect, Sylhet (0.93 WER even after fine-tuning), is consistently the hardest across all models, followed by the coastal/island varieties Sandwip and Old Dhaka. Central dialects (Khulna, Mymensingh, Narail, Noakhali) are the easiest, reaching 0.66–0.70 WER. The proposed model is best or tied on 14 of 15 dialects; only on Barishal does the larger zero-shot Whisper large-v3 win (0.82 vs. 0.87), indicating that model scale still matters for some dialects and motivating larger fine-tuned backbones as future work.

7.3 CS vs. BN_ONLY Difficulty Gap

Table 9 reports WER on CS vs. BN_ONLY clips. The whole-clip gap is small (within ± 0.12): because code-switched clips are predominantly Bengali, code-switching does not substantially inflate overall clip WER.

Rather, the difficulty of code-switching is concentrated in the English tokens themselves. As Table 7 shows, English-word survival is near zero for monolingual and BN_ONLY models and English-word WER approaches 1.0, even though overall clip WER barely changes. Code-switching difficulty is therefore real but localised—it manifests as the systematic loss of English words, not as a higher overall clip WER. Fine-tuning on code-switched data is what recovers these English tokens (CS-F1 and EN-survival in Tables 5 and 7).

7.4 Pronunciation Normalisation Impact

Table 10 shows the reduction in WER when pronunciation variant normalisation is applied to the evaluation references and hypotheses.

The WER reduction from normalisation is larger for zero-shot models ([X.X] points) than for fine-tuned models ([X.X] points). This suggests that fine-tuning on BanglaMix data helps models learn consistent handling of English loanword orthography, reducing dependency on post-hoc normalisation. The normalisation lexicon covers 271 canonical forms (649 spelling variants) spanning technology, academic, finance, lifestyle, and high-frequency CS adverbs (seriously, actually, basically), which are particularly common in the comedy and vlog domains of our dataset.

7.5 LLM Judge Validation (Dataset Quality Control)

Annotation pipeline statistics. Table 11 shows the outcome of the annotation pipeline. The high-confidence clips are auto-accepted in Step 4b; the borderline clips that Whisper is uncertain about are adjudicated by the single-model Qwen2.5 judge in Step 4c.

Judge decisions. Of the 8,712 borderline clips, the Qwen2.5 judge accepted 1,727 (19.8%) and rejected 6,985 (80.2%); no clip was left undecided. The high rejection rate is expected: these are exactly the low-confidence clips Whisper flagged, and the judge serves as a stricter second gate that recovers a useful minority while discarding noise. The recovered 1,727 clips are added to the 39,772 auto-accepted clips, yielding the pre-filter annotated pool that is passed to dataset construction.

Dimension-level rejection. The judge rejected clips primarily on the Bengali validity ([XX.X%] rejection rate) and not-noise ([XX.X%]) dimensions. CS naturalness and dialect authenticity had lower rejection rates ([XX.X%] and [XX.X%] respectively), indicating that borderline clips failing Whisper’s confidence threshold are more often noise artefacts than content quality problems.

7.6 LLM-Ensemble Evaluation

Beyond string-level metrics, we evaluate ASR hypotheses with a three-model LLM-as-a-judge ensemble (Mistral-7B [Jiang et al., 2023], LLaMA-3.1-8B [Dubey et al., 2024], and Gemma [Gemma Team et al., 2024]), each run locally via Ollama. Every model scores the hypothesis against the reference on three 0–5 dimensions—semantic correctness, code-switch preservation, and dialectal naturalness—and the ensemble reports the mean score per dimension together with an inter-model agreement rate. This metric captures meaning- preserving variation (e.g. miting/meeting) and silent dropping of English code-switch words, which WER penalises or misses. Table 12 reports ensemble scores for the proposed system and the strongest baselines.

Silver-standard quality. To validate the quality of silver-standard transcripts, we manually transcribed a random sample of 200 CS clips from the test set and computed human-verified WER against the Whisper-generated labels. Human-verified WER (silver vs. human) is [XX.X%], indicating that Whisper large-v3 with forced Bengali decoding produces transcripts of sufficient quality to serve as training labels. This is consistent with Gandhi [2023] for other low-resource languages.

7.7 Statistical Significance

All WER improvements of the proposed system (Whisper FT ALL) over the baselines are statistically significant at $p < 0.001$ (paired bootstrap, $N = 10,000$). The 95% confidence interval for the proposed system’s overall WER is [0.79, 0.81], which does not overlap the interval of any baseline (e.g. zero-shot Whisper large-v3, [0.90, 0.92]). The effect sizes are statistically robust but numerically small (Cohen’s $d = 0.09$ vs. large-v3), as expected when comparing models that all operate in a high-error regime on this challenging benchmark.

8. Conclusion

This work presented **BanglaMix**, the first large-scale dataset and benchmark for dialectal Bengali–English code-switching ASR. The dataset comprises 51,857 clips totalling 164.74 hours across 15 regional dialects of Bangladesh, collected from YouTube and Facebook using dialect-name search queries targeting content genres where natural CS is

most prevalent. An automated pipeline based on WebRTC VAD segmentation, Whisper silver-standard transcription, and a single-model LLM judge (Qwen2.5) for borderline clip validation enables large-scale annotation without any human transcribers. For evaluation, a separate three-model LLM ensemble (Mistral-7B, LLaMA-3.1-8B, Gemma) rates hypotheses beyond surface WER.

This work benchmarked three ASR architectures—Whisper, Meta MMS, and wav2vec2-XLS-R—under zero-shot and fine-tuned conditions. Key findings are:

1. Zero-shot multilingual models substantially underperform on dialectal CS speech (≥ 0.91 WER), even state-of-the-art Whisper large-v3.
2. Dialect-only fine-tuning (Ablation A) and code-switch-only fine-tuning (Ablation B) trade off Bengali vs. English recognition: BN_ONLY barely reproduces English (EN survival 0.18), while CS_ONLY best preserves it (0.47) at the cost of overall accuracy. Training on all data gives the best balance.
3. The proposed Whisper FT ALL model achieves **0.80 WER / 0.49 CER** overall (0.82 WER on CS clips), a 25% relative improvement over its zero-shot counterpart and a statistically significant gain over every baseline ($p < 0.001$).
4. Absolute error remains high: dialectal Bengali has no standardised orthography and labels are silver-standard, so BanglaMix is best viewed as a challenging open benchmark rather than a solved task.
5. Per-dialect analysis reveals that the most phonologically distinct dialect, Sylhet, is the hardest across all models, suggesting future work should prioritise targeted data collection for such varieties.

Limitations. BanglaMix relies on silver-standard transcripts from Whisper, which may contain systematic errors—particularly for words spoken in strong dialect with no standard Bengali equivalent. The code-switched proportion ($\sim 7\%$) reflects the natural prevalence of CS in our collected content; more aggressive query targeting or domain selection could increase this. The dataset is limited to audio from social media, which may not generalise to formal domains (news, education, healthcare).

Future work. Future directions include: (1) expanding to dialects with no current coverage (Jessore, Faridpur, Cox’s Bazar); (2) domain extension to medical and academic code-switching; (3) developing dialect identification as an auxiliary task to improve ASR; (4) human post-correction of the highest-impact clips using active learning to maximise annotation efficiency.

Data and code availability. The BanglaMix dataset, all model checkpoints, and the complete data collection and evaluation pipeline are released at [repository URL] under a CC-BY 4.0 licence (audio from publicly available sources; transcripts under CC0).

Acknowledgements

[Acknowledgements will be added here.] This work was supported by [funding agency / grant number].

তথ্যসূত্রসমূহ

- Tanvir Ahmed and Md Mahadi Hasan. Sylheti speech data for low-resource ASR. Technical Report, BUET, 2019.
- Md. Mahbub Ali, Md. Zakir Hossain Islam, and Md. Monirul Islam. Automatic recognition of spoken Bangla numerals using MFCC and DTW. In Proceedings of the International Conference on Advances in Electrical Engineering (ICAEE), 2015.
- Arun Babu, Changan Wang, Andros Tjandra, Kushal Lakhotia, Qiantong Xu, Naman Goyal, Kritika Singh, Patrick von Platen, Yatharth Saraf, Juan Pino, Alexei Baevski, Alexis Conneau, and Michael Auli. XLS-R: Self-supervised cross-lingual speech representation learning at scale. In Proc. Interspeech 2022, pages 2278–2282, 2022.
- Taylor Berg-Kirkpatrick, David Burkett, and Dan Klein. An empirical investigation of statistical significance in NLP. In Proceedings of the 2012 Joint Conference on Empirical Methods in Natural Language Processing and Computational Natural Language Learning (EMNLP-CoNLL), pages 995–1005. Association for Computational Linguistics, 2012.
- William Chan, Navdeep Jaitly, Quoc V. Le, and Oriol Vinyals. Listen, attend and spell: A neural network for large vocabulary conversational speech recognition. In Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pages 4960–4964, 2016.
- Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian, Jian Wu, Michael Zeng, and Furu Wei. WavLM: Large-scale self-supervised pre-training for full stack speech processing. IEEE Journal of Selected Topics in Signal Processing, 16(6):1505–1518, 2022.
- Qauser Ahmed Chowdhury and Mahfuzur Rahman Khan. Dialectal variation in Bengali: A sociolinguistic overview of regional speech in Bangladesh. International Journal of Language and Linguistics, 2(6):1–9, 2014.
- Abhinav Diwan, Vandan Mujadia R, Rakesh Saluja, Kalika Bali, Monojit Choudhury, and Sunayana Sitaram. MUCS 2021: Multilingual and code-switching ASR challenges for low resource Indian languages. In Proc. Interspeech 2021, pages 2446–2450, 2021.
- Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, et al. The Llama 3 herd of models. arXiv preprint arXiv:2407.21783, 2024.
- Jinlan Fu, See-Kiong Ng, Zhengbao Jiang, and Pengfei Liu. GPTScore: Evaluate as you desire. arXiv preprint arXiv:2302.04166, 2023.
- Sanchit Gandhi. Fine-tuning whisper for low-resource speech recognition. HuggingFace Blog, 2023. URL <https://huggingface.co/blog/fine-tune-whisper>.

Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, Thomas Mesnard, Bobak Shahriari, Alexandre Ramé, Johan Ferret, Peter Liu, Pouya Tafti, Abe Friesen, Michelle Casbon, Sabela Ramos, Ravin Kumar, et al. Gemma 2: Improving open language models at a practical size. arXiv preprint arXiv:2408.00118, 2024.

Google. WebRTC voice activity detector, 2012. URL <https://webrtc.org>.

Alex Graves, Santiago Fernández, Faustino Gomez, and Jürgen Schmidhuber. Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks. In Proceedings of the 23rd International Conference on Machine Learning (ICML), pages 369–376. ACM, 2006.

Awni Hannun, Carl Case, Jared Casper, Bryan Catanzaro, Greg Diamos, Erich Elsen, Ryan Prenger, Sanjeev Satheesh, Shubho Sengupta, Adam Coates, and Andrew Y. Ng. Deep speech: Scaling up end-to-end speech recognition. arXiv preprint arXiv:1412.5567, 2014.

Md Mahadi Hasan, Md Saiful Islam, Tanvir Ahmed, and Firoj Alam. Ben-10: A large-scale multi-dialect Bengali speech corpus for automatic speech recognition. In Proceedings of the 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (AACL), 2025.

M. A. Hasnat, Jabir Mowla, and S. M. Saif. Isolated and continuous Bangla speech recognition: Implementation, performance and pit-falls. In Proceedings of the International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS), pages 452–455, 2007.

HuggingFace. Evaluate: A HuggingFace library for reproducible evaluation, 2022. URL <https://github.com/huggingface/evaluate>.

Md. Saiful Islam and Firoj Alam. A spoken corpus of Chittagonian (Chatgaiya) Bengali. Technical Report, 2022.

Md. Saiful Islam, Moin Ahsan, and Tamim Iqbal. End-to-end attention-based Bangla speech recognition using sequence-to-sequence models. In Proceedings of the International Conference on Natural Language, Signal and Speech Processing (ICNLSSP), 2019.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Léo Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. Mistral 7B. arXiv preprint arXiv:2310.06825, 2023.

Seungwhan Kim, Seungwhan Moon, Paul Crook, Stephen Roller, Anastasia Spiliopoulou, Yoav Artzi, and William Yang Wang. Can LLMs serve as judges for dialogue evaluation? assessing spoken dialogue system outputs. In Proc. Interspeech 2023, pages 3789–3793, 2023.

- Oddur Kjartansson, Manuk Aznaveh, Lara Helgadottir, Sigrún Mollberg, Christopher Cieri, Símon Kjaran, and Börkur Sigurbjörnsson. Crowd-sourced speech corpora for Javanese, Sundanese, Sinhala, Nepali, and Bengali and software tooling for the Bengali ASR system. In *Proceedings of the Interspeech 2018 Workshop on Spoken Language Technologies for Under-Resourced Languages (SLTU)*, 2018. URL <https://openslr.org/37/>.
- Quentin Lhoest, Albert Villanova del Moral, Yacine Jernite, Abhishek Thakur, Patrick von Platen, Suraj Patil, Julien Chaumond, Mariama Drame, Julien Plu, Lewis Patry, Angelina McMillan-Major, Philipp Schmid, Sylvain Gugger, Clement Delangue, Theo Matussiere, Lysandre Debut, Stas Bekman, Pierric Cistac, Thibault Goehringer, Victor Mustar, Francois Lagunas, Alexander Rush, and Thomas Wolf. Datasets: A community library for natural language processing. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP)*, pages 175–184. Association for Computational Linguistics, 2021.
- Andrew C. Morris, Viktoria Maier, and Phil Green. jiwer: Evaluate your speech-to-text system with WER, MER, WIL, WIP and CER, 2004. URL <https://github.com/jitsi/jiwer>.
- Vineel Pratap, Andros Tjandra, Bowen Shi, Paden Tomasello, Arun Babu, Sayam Xu, Alexei Baevski, Alexis Conneau, Ronan Collobert, Yatharth Saraf, and Michael Auli. Scaling speech technology to 1,000+ languages. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*. PMLR, 2024.
- Qwen Team. Qwen2.5 technical report. arXiv preprint arXiv:2412.15115, 2024.
- Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, pages 28492–28518. PMLR, 2023.
- Sunayana Sitaram, Khyathi Raghavi Chandu, Sree Krishna Rallabandi, and Alan W. Black. A survey of code-switched speech and language processing. arXiv preprint arXiv:1904.00784, 2019.
- Yidong Wang, Zhuohao Yu, Zhengran Zeng, Linyi Yang, Cunxiang Wang, Hao Chen, Chaoya Jiang, Rui Xie, Jindong Wang, Xing Xie, Wei Ye, Shikun Zhang, and Yue Zhang. PandaLM: An automatic evaluation benchmark for LLM instruction tuning optimization. arXiv preprint arXiv:2306.05087, 2023.
- Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander M. Rush. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP)*, pages 38–45. Association for Computational Linguistics, 2020.

Emre Yilmaz, Henk van den Heuvel, and David A. van Leeuwen. Investigating acoustic and linguistic context effects in code-switching ASR. In Proc. Interspeech 2016, pages 3557–3561, 2016.

yt-dlp contributors. yt-dlp: A feature-rich command-line audio/video downloader, 2021. URL <https://github.com/yt-dlp/yt-dlp>.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In Advances in Neural Information Processing Systems (NeurIPS), volume 36, 2023.

সারণী 2: Comparative summary of representative Bengali / code-switching ASR systems and datasets. BanglaMix is the first to combine multi-dialect Bengali with English code-switching. (CS = code-switching; ✓ present, ✗ absent; “–” not applicable.)

Work	Method / Model	Variety	Dialects	CS	Hours	Key limitation
Hasnat et al. [2007]	GMM-HMM	Standard BN	1	✗	–	Lexicon-dependent; WER >50%; no OOV
Ali et al. [2015]	DNN-HMM	Standard BN	1	✗	–	Lexicon-dependent; standard BN only
Islam et al. [2019]	End-to-end (CTC/attn.)	Standard BN	1	✗	–	Standard BN only; no dialect/CS
Kjartansson et al. [2018]	Corpus (OpenSLR-37)	Standard BN	1	✗	100	Read speech; no dialect, no CS
Diwan et al. [2021]	CS challenge corpus	Standard BN	1	✓	[~5]	Standard Dhaka BN; no dialect
Hasan et al. [2025]	Whisper FT (Ben-10)	Dialectal BN	10	✗	78	No code-switching
Babu et al. [2022]	wav2vec2-XLS-R (SSL)	Multilingual	–	✗	–	Standard BN via transfer; no dialect/CS
Radford et al. [2023]	Whisper (weak sup.)	Multilingual	–	✗	–	~70% WER on unseen BN dialects
BanglaMix (ours)	Whisper/CTC FT + benchmark	Dialectal BN	15	✓	[~165]	First multi-dialect BN + CS dataset

সারণী 3: BanglaMix dialect statistics after segmentation (Step 2).

Dialect	Region	Clips	Hours
Sylhet	Northeast Bangladesh + UK diaspora	9,665	31.85
Old_Dhaka	Historic Dhaka city	5,767	17.84
Sandwip	Island, Chittagong coast	5,744	16.31
Barishal	South-central Bangladesh	4,048	13.57
Khulna	Southwest Bangladesh	3,540	11.40
Chittagong	Southeast Bangladesh	3,661	11.16
Kishoreganj	Central Bangladesh	3,038	9.52
Comilla	Southeast Bangladesh	2,724	9.39
Narail	Southwest Bangladesh	2,714	8.93
Rangpur	Northwest Bangladesh	2,650	8.67
Narsingdi	Central Bangladesh	2,150	6.63
Mymensingh	North-central Bangladesh	1,831	5.50
Habiganj	Northeast Bangladesh	1,694	5.28
Tangail	Central Bangladesh	1,396	4.77
Noakhali	Southeast Bangladesh	1,235	3.93
Total	15 dialects, 9 districts	51,857	164.74

সারণী 4: BanglaMix train/dev/test split statistics, after leakage-safe source-level splitting and transcript quality filtering. The training set includes audio augmentation; dev/test are unaugmented.

Split	Clips	Hours	CS clips	CS%
Train	60,936	188.4	4,485	7.4%
Dev	2,806	9.1	166	5.9%
Test	4,088	13.2	176	4.3%
Total	67,830	210.8	4,827	7.1%

সারণী 5: Main results on the BanglaMix test set. WER/CER: lower is better. CS-F1: higher is better. **Bold green** = best overall. † = statistically significant improvement over Whisper large-v3 zero-shot ($p < 0.05$, paired bootstrap).

Model	WER↓	CER↓	WER-BN↓	WER-EN↓	MIX-ER↓	WER-CS↓	CS-F1↑
Zero-shot baselines (no fine-tuning on BanglaMix)							
Whisper-small (zero-shot)	1.07	1.02	1.07	1.26	1.15	1.02	0.00
Whisper large-v3	0.91	0.76	0.92	1.08	0.98	0.80	0.14
MMS-1B	0.99	0.57	0.99	1.11	—	0.94	0.04
wav2vec2-XLS-R-BN	0.96	0.62	0.96	1.03	—	0.94	0.04
CTC model fine-tuned on BanglaMix							
wav2vec2 (fine-tuned)†	0.86	0.49	0.86	1.00	0.97	0.88	0.14
Whisper fine-tuned on BanglaMix — ablation							
Whisper FT BN_ONLY (A)†	0.80	0.49	0.80	1.00	0.94	0.82	0.01
Whisper FT CS_ONLY (B)	0.88	0.55	0.89	0.76	0.82	0.81	0.28
Whisper FT ALL (Proposed)†	0.80	0.49	0.80	0.93	0.91	0.82	0.19

সারণী 6: Per-dialect WER for the proposed model (Whisper FT ALL).

Dialect	Clips	WER↓	CER↓
Sylhet	1,111	0.93	0.65
Old_Dhaka	360	0.78	0.53
Sandwip	176	0.92	0.67
Barishal	282	0.87	0.54
Chittagong	198	0.79	0.46
Khulna	120	0.66	0.39
Kishoreganj	743	0.75	0.36
Comilla	157	0.78	0.42
Narail	464	0.70	0.41
Rangpur	165	0.79	0.51
Narsingdi	60	0.72	0.39
Mymensingh	56	0.70	0.35
Habiganj	75	0.90	0.52
Tangail	51	0.74	0.39
Noakhali	70	0.70	0.36
Overall	4,088	0.80	0.49

সারণী 7: Error type breakdown and English word survival rate. S/D/I = substitution/deletion/insertion as fraction of reference words. EN-Surv = fraction of English reference words found in hypothesis.

Model	S	D	I	EN-Surv↑
Whisper large-v3 (0-shot)	0.31	0.48	0.12	0.65
Whisper-small (0-shot)	0.25	0.75	0.07	0.12
MMS-1B	0.66	0.13	0.19	0.00
wav2vec2 (zero-shot)	0.52	0.36	0.08	0.00
wav2vec2 FT	0.52	0.23	0.10	0.00
Whisper FT BN_ONLY (A)	0.45	0.21	0.14	0.01
Whisper FT CS_ONLY (B)	0.53	0.19	0.16	0.37
Whisper FT ALL	0.45	0.21	0.14	0.13

সারণী 8: Per-dialect WER across representative models. **Bold green** = best per dialect. Fine-tuning helps almost every dialect, but the larger zero-shot model retains an edge on Barishal.

Dialect	large-v3 (zs)	wav2vec2 (FT)	Whisper ALL
Sylhet	1.09	0.93	0.93
Old_Dhaka	0.97	0.86	0.78
Sandwip	1.09	0.94	0.92
Barishal	0.82	0.87	0.87
Chittagong	0.80	0.87	0.79
Khulna	0.81	0.75	0.66
Kishoreganj	0.79	0.83	0.75
Comilla	0.88	0.87	0.78
Narail	0.82	0.79	0.70
Rangpur	1.00	0.85	0.79
Narsingdi	0.79	0.75	0.72
Mymensingh	0.86	0.83	0.70
Habiganj	0.93	0.95	0.90
Tangail	0.80	0.83	0.74
Noakhali	0.80	0.83	0.70

সারণী 9: WER on CS vs. BN_ONLY clips. Gap = WER-CS – WER-BN.

Model	WER-CS	WER-BN	Gap
Whisper large-v3 (0-shot)	0.80	0.92	−0.12
Whisper-small (0-shot)	1.02	1.07	−0.05
MMS-1B	0.94	0.99	−0.05
wav2vec2 (fine-tuned)	0.88	0.86	+0.02
Whisper FT CS_ONLY (B)	0.81	0.89	−0.08
Whisper FT ALL	0.82	0.80	+0.02

সারণী 10: Impact of pronunciation variant normalisation on WER. WER_{raw} uses unmodified transcripts; WER_{norm} maps phonetic and script variants of English loanwords to a canonical form before scoring.

Model	WER_{raw}	WER_{norm}	Reduction
Whisper large-v3 (0-shot)	[0.XX]	[0.XX]	[−X.X%]
Whisper-small (0-shot)	[0.XX]	[0.XX]	[−X.X%]
wav2vec2 FT + LM	[0.XX]	[0.XX]	[−X.X%]
Whisper FT ALL	[0.XX]	[0.XX]	[−X.X%]

সারণী 11: Annotation outcomes. Step 4c counts are from the Qwen2.5 judge over the borderline clip set. Counts are pre-augmentation and prior to the final transcript quality filtering applied at dataset-build time (Table 4).

Stage	Clips	Note
Step 4b auto-accept (prob < 0.30)	39,772	auto-accepted
Step 4b auto-reject (prob ≥ 0.60)	[XX,XXX]	auto-rejected
Step 4c Qwen2.5 judge — ACCEPT	1,727	19.8% of judged
Step 4c Qwen2.5 judge — REJECT	6,985	80.2% of judged
Step 4c Qwen2.5 judge — UNCERTAIN	0	excluded
Annotated pool (pre-filter)	41,499	input to Step 5

সারণী 12: LLM-ensemble evaluation of ASR outputs (mean of three judges, 0–5; higher is better). Agr. = mean inter-model agreement.

Model	Semantic	CS-Pres.	Dialect	Agr.
Whisper large-v3 (0-shot)	[X.X]	[X.X]	[X.X]	[0.XX]
Whisper-small (0-shot)	[X.X]	[X.X]	[X.X]	[0.XX]
Whisper FT ALL	[X.X]	[X.X]	[X.X]	[0.XX]